

Intergalactic Riksbanken Chip Authenticator - Project Summary

Group 8

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Intergalactic Riksbanken Chip Authenticator

Project Summary

Authors: Group 8

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Course: STB600 Final Project 2025

Date: December 14, 2025

■ Project Complete!

■ *Deliverables*

- **Source Code** - All three operating modes implemented
- **Documentation** - Complete technical design document
- **Screenshots** - 7 demonstration images captured
- **User Guides** - Quick start and detailed usage instructions
- **Design Report** - 30-page comprehensive project document

■ Project Structure

chip_system/	# Main project directory
■■■ launcher.py	# ■ Main entry point (all modes)
■■■ main.py	# Conveyor belt simulator
■■■ camera_main.py	# Real-time camera detection
■■■ game.py	# Interactive manual testing
■■■ capture_screenshots.py	# Documentation utility
■	
■■■ assets/	# Chip template images
■ ■■■ gold.png	# Gold chip with green background
■ ■■■ silver.png	# Silver chip
■ ■■■ bronze.png	# Bronze chip

```

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■■■ docs/                                # Documentation folder
■   ■■■ images/                          # ■ Project screenshots
■   ■   ■■■ 01_empty_conveyor.png
■   ■   ■■■ 02_multiple_chips.png
■   ■   ■■■ 03_gold_chip.png
■   ■   ■■■ 04_silver_chip.png
■   ■   ■■■ 05_bronze_chip.png
■   ■   ■■■ 06_fake_chip.png
■   ■   ■■■ 07_full_simulation.png
■   ■■■ SCREENSHOTS.md                  # Screenshot documentation
■   ■■■ CAMERA_USAGE.md                 # Camera system guide
■   ■■■ MIGRATION.md                    # Cleanup notes
■
■■■ README.md                           # Main documentation
■■■ DESIGN_DOCUMENT.md                  # ■ Complete technical report
■■■ QUICKSTART.md                       # Fast reference guide
■■■ CHANGELOG.md                        # Version history
■■■ COMMIT_SUMMARY.md                  # Git commit preparation
■■■ CONTRIBUTING.md                     # Contribution guidelines
■■■ LICENSE                             # MIT License
■■■ requirements.txt                     # Python dependencies
■■■ .gitignore                           # Git ignore rules

```

■ Three Operating Modes

1. Simulator Mode (``python main.py``)

- Conveyor belt with automatic chip spawning
- 60 FPS real-time simulation
- Perfect for algorithm testing

2. Camera Mode (``python camera_main.py``)

- Interactive color calibration
- Real-time chip detection
- Supports webcam and Basler cameras
- 30+ FPS processing

3. Interactive Game (``python game.py``)

- Manual chip spawning (keys 1/2/3)
- Static display for testing
- Grid background for placement
- Instant value feedback

Unified Launcher (``python launcher.py``)

- Single entry point for all modes
- Interactive menu system
- Error handling and recovery

■ System Capabilities

Chip Authentication

- **Gold Chips:** Yellow, value = digits × 10
- **Silver Chips:** Gray, value = digits
- **Bronze Chips:** Orange, value = digits multiplied
- **Fake Chips:** Red, zero value

Technical Features

- ■ HSV color space detection
- ■ Alpha channel transparency
- ■ Real-time statistics tracking
- ■ 60 FPS simulator, 30+ FPS camera
- ■ Green background removal
- ■ Automatic value calculation

■ Screenshots Captured

All screenshots are located in docs/images/:

01_empty_conveyor.png - Initial state with green belt

02_multiple_chips.png - Multiple chips in motion

03_gold_chip.png - Gold chip detection

04_silver_chip.png - Silver chip detection

05_bronze_chip.png - Bronze chip detection

06_fake_chip.png - Fake chip identification

07_full_simulation.png - Complete system view

View all screenshots with descriptions in docs/SCREENSHOTS.md

■ Documentation Files

User Documentation

- **README.md** (140 lines) - Complete project overview
- **QUICKSTART.md** (50 lines) - Fast reference
- **docs/SCREENSHOTS.md** (100 lines) - Visual demonstrations

Technical Documentation

- **DESIGN_DOCUMENT.md** (700+ lines) - Complete technical design & project report
- Executive summary
- System architecture
- Algorithm descriptions
- Performance analysis
- Code structure
- Future enhancements

Development Documentation

- **docs/CAMERA_USAGE.md** (200+ lines) - Camera system guide
- **CONTRIBUTING.md** (80 lines) - Contribution guidelines
- **CHANGELOG.md** (40 lines) - Version history
- **COMMIT_SUMMARY.md** (150 lines) - Git preparation

■ Technical Stack

Component	Technology	Version
Language	Python	3.8+
Vision	OpenCV	4.8.0+
Computing	NumPy	1.24.0+

Color Space	HSV	OpenCV
Camera	Webcam/Basler	Any

■ Key Achievements

■ **Multi-Mode System:** Simulator, Camera, Game modes

■ **Real-Time Processing:** 30-60 FPS performance

■ **Adaptive Calibration:** Interactive color learning

■ **Complete Documentation:** 30+ page technical report

■ **Visual Demonstrations:** 7 professional screenshots

■ **Clean Architecture:** Modular, extensible design

■ **User-Friendly:** Intuitive controls and feedback

■ How to Run

Quick Start

```
# Navigate to project
cd chip_system

# Install dependencies
pip install -r requirements.txt

# Run launcher (recommended)
python launcher.py

# Select mode:
# 1 - Simulator
# 2 - Camera (with calibration)
# 3 - Interactive Game
```

Individual Modes

```
# Simulator only
python main.py

# Camera only
python camera_main.py

# Game only
```

python game.py

■ Project Report

The complete technical design document is available in:

■ DESIGN_DOCUMENT.md

Includes:

- System architecture diagrams
- Algorithm explanations with formulas
- Performance benchmarks
- Code structure analysis
- Testing & validation results
- Future enhancement roadmap

■ Group 8 Members

Name	Role	Contributions
Suneela	Team Member	System development
Sara	Team Member	System development
Abhishek	Team Member	System development

■ License

MIT License - Copyright (c) 2025 Group 8 (Suneela, Sara, Abhishek)

■ Project Status: COMPLETE ■

All deliverables completed:

- ■ Working software (3 modes)
- ■ Technical documentation
- ■ Project screenshots
- ■ User guides
- ■ Design report
- ■ Source code organization
- ■ Ready for submission

Intergalactic Riksbanken Chip Authenticator

Authenticating the future, one chip at a time ■

STB600 Final Project 2025

Group 8: Suneela, Sara, Abhishek